

Calangan, Ivan 1. WM221

1st ep
 1. 13, 11, False Hit 0 (23, 11, false)
 2. 23, 11, false Hit -1 Bust

2nd ep
 1. 21, 2, True Stand +1 Win

3rd episode
 1. 13, 11, false Hit 0 23, 11, false
 2. 23, 11, false Hit -1 Bust

4th episode
 11, 4, false Hit 0 19, 4, false
 19, 4, false Stand 0 19, 20, false
 19, 20, false Stand -1 Bust

5th episode
 19, 6, false Stand 0 19, 6, false
 19, 6, false Stand 0 19, 16, false
 19, 16, false Stand +1 Win

6th episode
 16, 3, false Hit 0 26, 3, false
 26, 3, false Hit -1 Bust

7th episode
 15, 6, false Hit 0 25, 6, false
 25, 6, false Stand 0 25, 26, false
 25, 26, false Stand -1 Bust

8th episode
 5, 10, false Hit 0 15, 10, true
 15, 10, true Hit 0 22, 10, true
 22, 10, true Hit -1 Bust

9th episode
 13, 2, false Hit 0 18, 2, false
 18, 2, false Stand 0 18, 22, false
 18, 22, false Stand +1 Win

10th
 12, 5, false Hit 0 20, 5, false
 20, 5, false Stand 0 20, 20, false
 20, 20, false Stand 0 tie



Monte Carlo

Step	State	Action	Reward	Next State
1	13, 2, take	Hit	6	18, 2, take
2	18, 2, take	Stand	0	18, 22, Take
3	18, 22, take	Stand	+1	Win

Step 1

State	Return	N(s)	old V(s)	New V(s)
13, 2, Take	+1	+1	0	0

a. 13, 2, Take

b. Number of visits: (0)

c. $V(13, 2, \text{Take}) = 0$

d. visit count: 1

e. Update value function = $0 + \frac{1}{1} (1 - 0) = 1$

Step 2

State	Return	N(s)	old V(s)	New V(s)
13, 2, take	+1	+1	0	+1
18, 2, take	+1	+1	0	0

a. 18, 2, take

b. Number of visits: (0)

c. $V(18, 2, \text{Take}) = 0$

d. visit count = 1

e. Update value function = $0 + \frac{1}{1} (1 - 0) = 1$

Step 3

State	Return	N(s)	old V(s)	New V(s)
13, 2, Take	+1	+1	0	+1
18, 2, Take	+1	+1	0	+1
18, 22, take	+1	+1	0	0

a. 18, 22, take

b. Number of visits: (0)

c. $V(18, 22, \text{Take}) = 0$

d. visit count = 1

e. Update value function = $0 + \frac{1}{1} (1 - 0) = 1$

State	Return	N(s)	old V(s)	New V(s)
13, 2, Take	+1	+1	0	+1
18, 2, Take	+1	+1	0	+1
18, 22, Take	+1	+1	0	+1

VICTORY

Temporal Difference

Step 1

Step	State s	Reward r	Next State s'	old V(s)	New
1	13, 2, False	0	18, 2, False	0	0
2					

a.) State = 13, 2, False

b.) Reward = 0

c.) Next State = 18, 2, False

d.) Update = $V(13, 2, \text{False}) = 0 + 0.5 (0 + V(18, 2, \text{False}) - 0) = 0$

Step 2

a.) State = 18, 2, False

b.) Reward = 0

c.) Next State = 18, 22, False

d.) Update = $V(18, 2, \text{False}) = 0 + 0.5 (0 + V(18, 22, \text{False}) - 0) = 0$

Step	State s	Reward r	Next State s	old V(s)	New
1	13, 2, False	0	18, 2, False	0	0
2	18, 2, False	0	18, 22, False	0	0

Step 3

a.) State = 18, 22, False

b.) Reward = 1

c.) Next State = None / Win

d.) Update = $V(18, 22, \text{False}) = 0 + 0.5 (1 + 0 - 0) = 0.5$

Step	State s	Reward r	Next State s	old	New
1	13, 2, False	0	18, 2, False	0	0
2	18, 2, False	0	18, 22, False	0	0
3	18, 22, False	+1	None / Win	0	+0.5

Step 4

a.) State = 13, 2, False

b.) Reward = 1

c.) Next State = None 18, 2, False

d.) Update = $V(13, 2, \text{False}) = 0 + 0.5 (1 + 0.5 - 0) = 0.75$

Step	State s	Reward r	Next State s	old	New
1	13, 2, False	+1	18, 2, False	0	+0.75
2	18, 2, False	0	18, 22, False	0	0
3	18, 22, False	+1	None / Win	0	+0.5

Step 5

a.) State = 18, 2, false

b.) Reward = +1

c.) Next State: 18, 22, false

d.) Update: $V(18, 2, \text{false}) = 0 + 0.5(1 + 0.75 - 0) = 0.875$

Step	State S	Reward r	Next State 's'	old	New
1	13, 2, false	+1	18, 2, false	0	+0.75
2	18, 2, false	+1	18, 22, false	0	+0.875
3	18, 22, false	+1	None / win	0	+0.5